

BaiZe Technical Report

Abstract—Large language models (LLMs) exhibit strong reasoning and tool-use skills, yet their computational footprint makes most agentic models impractical for edge, embedded, and cost-sensitive electronic design automation (EDA) deployments. We present BaiZe, a ≈ 2.2 -billion-parameter agentic language model built *from scratch* on a fixed compute budget around a *hybrid* state-space backbone. BaiZe adopts a sequential hybrid architecture—56 layers interleaving Mamba-2 state-space layers, four grouped-query-attention layers, and MLP layers—whose linear-time decoding is well suited to inference-heavy, cost-sensitive agents, and is trained with the NVIDIA NeMo recipe on Megatron-Core. The architecture is chosen through a matched from-scratch comparison against a dense transformer counterpart, in which the hybrid attains lower loss and faster inference with fewer parameters at a modest training-throughput cost. We then run a staged, budget-constrained hyper-parameter search (S1–S5) on $8\times H100$ over the final hybrid backbone, which selects a stable learning rate of 1×10^{-3} under a warmup–stable–decay schedule with an 86:10:4 annealing mixture, reaching a validation loss of 2.2054 after 20000 steps and 2.6739 ± 0.0469 across three seeds at 5000 steps. We release the complete recipe and training command to support the reproducible construction of efficient agentic small language models.

Index Terms—small language model, agentic LLM, hybrid architecture, state-space model, budget-constrained search, efficient inference

I. INTRODUCTION

Large language models have recently demonstrated agentic capabilities—planning, tool invocation, and long-horizon reasoning—that make them attractive cores for intelligent assistants spanning chip design, verification, and other domains of electronic design automation. However, state-of-the-art agent backbones are routinely 7B parameters or larger, demanding high-end accelerators and memory footprints that are incompatible with edge, embedded, and cost-sensitive deployments where inference latency and energy are first-class constraints. The constraint is sharper than raw cost in the settings we target: proprietary design artifacts—layouts, schematics and review judgements—cannot leave the premises, so an agentic core must be served locally; and because an agentic session accumulates context, what dominates serving cost is not prompt throughput but the memory that grows with the session. For a hybrid backbone this is architectural: only the four attention layers carry a session-growing state, while the Mamba-2 layers hold a constant-size recurrent one.

Recent small language models (SLMs, < 2 B) have closed much of the gap with larger counterparts on single-turn benchmarks, but building such a model *from scratch* remains an under-documented engineering problem. Every choice—architecture, optimizer, learning-rate schedule, data mixture—interacts with every other, and a naive grid search is prohibitively expensive. Our goal is therefore twofold: (i) to construct, from scratch and on a strictly limited compute

budget, an end-to-end multimodal agentic system—an efficient hybrid language core paired with a custom vision encoder, aligned into a semiconductor-domain agent—and (ii) to expose the concrete, reproducible decisions that lead to it.

a) *BaiZe.*: We present BaiZe, an end-to-end multimodal agentic system assembled from scratch across five stages: (i) pre-training a ≈ 2.2 B-parameter hybrid (state-space + attention) LLM, (ii) post-training it for agentic capability, (iii) pre-training a custom vision encoder, (iv) aligning the two into a multimodal model, and (v) adapting the result to the semiconductor domain. Three design pillars organise our work:

- **Architecture.** A sequential hybrid language backbone—56 layers interleaving Mamba-2 state-space layers, four grouped-query-attention layers, and MLP layers—giving linear-time decoding for inference-heavy agents, paired with a from-scratch vision encoder.
- **Reduced-horizon proxy search.** We exploit short, cheap proxy runs to rank candidate hyper-parameters, so that the ranking discovered at small scale carries over to the full model without re-tuning at scale.
- **Budget-constrained stage-wise pipeline.** Each of the five stages is trained under a hard GPU-hour budget, with a staged search protocol (S1–S5) for the language backbone and a single frozen evaluation protocol shared by the later stages.

We hope BaiZe serves as a minimal, reproducible blueprint for building efficient agentic SLMs from scratch, and that our search methodology is of practical value to cost-bounded training efforts in the EDA and embedded-AI communities.

II. BACKGROUND AND RELATED WORK

A. Small Language Models and Agentic Reasoning

Parameter-efficient SLMs (< 2 B) have recently narrowed the gap with larger models on mathematical and commonsense reasoning. MiniCPM [1] introduced scalable training strategies and domain-weighted data mixing for small models; Qwen [2] and Gemma [3] demonstrate that strong code- and math-enriched corpora can lift small models substantially. Most of these efforts, however, treat the final trained model as the deliverable and report comparatively little about the *search* that produced it. Complementarily, agentic instruction-tuning datasets (e.g., tool-use and multi-step reasoning corpora) [4] have been shown to elicit planning and tool-invocation behaviour, yet are rarely paired with a from-scratch pre-training recipe for the 1B regime.

B. Hybrid and State-Space Architectures

State-space models (SSMs) such as Mamba [5] and Mamba-2 [6] replace full attention with a recurrent scan update, of-

fering linear-time decoding that suits inference-bound deployments. Hybrid designs interleave a small number of attention layers among SSM/MLP layers—notably Nemotron-H [7] and Hymba [8]—to recover long-range context while retaining fast decode. Released Megatron-Core variants of these families are scarce below 2B, leaving their small-scale trade-offs under-documented; our controlled 2B comparison (Section IV-A) addresses precisely this gap.

C. Multimodal Vision-Language Models

Vision-language models connect a vision encoder to a language model through a lightweight projector, enabling end-to-end visual understanding and instruction following. LLaVA [9] established visual instruction tuning—aligning a vision encoder to a frozen LLM via a linear projection followed by instruction fine-tuning—as a scalable recipe for open multimodal assistants. LLaVA-OneVision [10] consolidates this line into a single open model spanning single-image, multi-image, and video tasks through a staged curriculum: stage-1 projector pre-training, stage-1.5 mid-training, and stage-2 instruction tuning. BaiZe follows this curriculum in stage (iv) (Section VII), substituting a from-scratch vision encoder (stage iii) for the frozen off-the-shelf encoder.

D. LLMs for Electronic Design Automation

A growing line of work applies LLM agents to chip design tasks such as register-transfer-level generation, verification, and bug triage [11], [12]. These applications motivate a cheap, always-on, agentic multimodal core that can run per-team or per-engineer; our work supplies the efficient multimodal backbone such systems require.

III. ARCHITECTURE AND PIPELINE OVERVIEW

BaiZe is an end-to-end multimodal agentic system assembled in five stages: (i) from-scratch pre-training of a hybrid language backbone, (ii) post-training of that backbone (instruction tuning and reinforcement learning to strengthen agentic capability), (iii) pre-training of a custom vision encoder, (iv) alignment of the two into a multimodal large language model, and (v) semiconductor-domain adaptation via reinforcement learning. This section fixes the shared scaffolding—the two backbones, the overall pipeline, and the global training environment—before each stage is detailed in its own section.

A. Language Backbone

The language backbone is a sequential hybrid model [6], [7]: 56 transformer layers interleave Mamba-2 state-space layers, four grouped-query-attention (GQA) layers, and MLP layers. The layer allocation follows the Nemotron-H sequential hybrid convention—four attention layers among the 56, the remainder Mamba-2 and MLP—as provided by Megatron-Core’s Mamba-hybrid recipe, whose 56-layer pattern we reuse verbatim while halving only the hidden and FFN widths to reach the ≈ 2 B scale. BaiZe is not a Nemotron-H model: it is an order of magnitude smaller, contains no mixture-of-experts layers, and is pre-trained from scratch on an independent

corpus under a fixed GPU-hour budget. Each Mamba-2 layer uses 8 state-space groups; the four attention layers use 16 query heads and 4 query groups (GQA) with untied word embeddings. The full configuration is listed in Table I; the model totals ≈ 2.22 B parameters.

TABLE I
BAIZE LANGUAGE-BACKBONE CONFIGURATION (MAMBA2-HYBRID).

Hyper-parameter	Value
Transformer layers	56
Architecture	hybrid (Mamba-2 + GQA + MLP)
Attention layers	4 (interleaved)
Hidden size	2048
FFN hidden size	8192
Attention heads (Q)	16
Query groups (GQA)	4
Mamba-2 state-space groups	8
Vocabulary size	129,281
Sequence length	4094
Activation	SiLU / SwiGLU
Normalisation	RMSNorm
Precision	BF16
Parameters	≈ 2.22 B

B. Vision Encoder

The vision encoder is pre-trained from scratch in stage (iii) and is selected from six iso-parameter candidates at the 500–600 M scale; Section VI reports the comparison and its outcome.

C. Pipeline and Global Environment

Table II summarises the five-stage pipeline. All stages run on $8 \times$ NVIDIA H100 GPUs (CUDA 12.8, PyTorch 2.8.0, Python 3.10); fused kernels, selective activation recomputation, and cross-entropy loss fusion are enabled where applicable, with `tensor_model_parallel_size = 1` and BF16 precision throughout.

TABLE II
BAIZE FIVE-STAGE PIPELINE OVERVIEW.

Stage	Task	Status	Section
(i)	LM pre-training	complete	IV
(ii)	Post-training (SFT, RLVR)	design	V
(iii)	Vision encoder	complete	VI
(iv)	MLLM alignment	design	VII
(v)	Domain adapt. (RLHF)	design	VIII

IV. LLM PRETRAINING

Stage (i) pre-trains the hybrid language backbone (Section III-A) from scratch under a fixed budget. This section covers the architecture-selection rationale, the training recipe, and the resulting measurements; the backbone configuration itself is given in Table I.

A. Architecture Selection Rationale

We adopt the hybrid backbone on three grounds. **(i) Inference efficiency.** State-space layers decode with constant-time recurrence per token, while the sparse attention layers retain long-range context; this directly targets the inference-heavy, cost-sensitive deployments that motivate BaiZe. **(ii) Controlled comparison.** A matched from-scratch comparison at the $\approx 2\text{B}$ scale between a dense transformer (MiniCPM5-2B) and Mamba2-hybrid—trained under identical data, seed, and GPU footprint—favours the hybrid, which converges to a lower loss with fewer parameters ($\approx 2.22\text{B}$ vs. 2.51B , -11.6%) and decodes faster under our direct-drive harness, at a $\sim 24\%$ training-throughput cost (Table IV). **(iii) Feasibility.** Megatron-Core ships a Mamba-hybrid provider whose 56-layer interleaving pattern we reuse verbatim, halving only the hidden and FFN widths to reach the $\approx 2\text{B}$ scale, making the backbone directly executable under the NeMo recipe without custom kernels.

B. Training Framework and Reduced-Horizon Search

We train with the NVIDIA NeMo pre-training entry point, driving the Megatron-Core [13] `NVIDIAMambaHybridModelProvider2B` provider directly with `tensor_model_parallel_size = 1`. To keep the entire sweep within budget, we run all hyper-parameter searches at a reduced iteration count and treat the resulting short-horizon ranking as predictive of the full-model optimum. Fused kernels, selective activation recomputation, and cross-entropy loss fusion are enabled to raise throughput.

C. Learning-Rate Schedule

BaiZe trains under a warmup–stable–decay–style learning-rate schedule whose hyper-parameters—warmup length, stable learning rate, and the decay ratio—are themselves swept in Section IV-D (cosine decay is kept as an alternative). The remaining schedule and batch-size settings (global batch size, sequence length, packing) are held fixed across configurations and re-measured on the final backbone, with figures reported where applicable.

D. Budget-Constrained Search Protocol

Because a full grid is infeasible under a fixed GPU budget, we adopt a staged, single-variable-at-a-time protocol:

- **S1 (baseline).** Reproduce the Mamba2-hybrid reference configuration (stable LR 3×10^{-4} , WSD) to establish a loss/throughput baseline.
- **S2 (learning-rate sweep).** Sweep the stable learning rate over seven points from 2×10^{-4} to 1×10^{-3} with WSD held fixed.
- **S3 (schedule axes).** On the winning learning rate, perturb one variable at a time—schedule family (WSD vs. cosine), decay ratio, warmup length, and `min_lr`.
- **S4 (annealing data mix).** Ablate the annealing data blend (pure Ultra-FineWeb-L3 vs. adding code and math) under the winning configuration.

- **S5 (long run + seeds).** Retrain the winning configuration from scratch over a 20000-step horizon, plus a three-seed short-horizon reproducibility check.

At every stage only a single variable changes relative to the running best, isolating each factor’s contribution while keeping total GPU-hours within budget. The architectural search space is validated up front: the hybrid (Mamba2-hybrid) and dense (MiniCPM5-2B) candidates are compared head-to-head from scratch (Section IV-A), which selects the hybrid backbone before the hyper-parameter search begins.

E. Data

Pre-training uses the Ultra-FineWeb-L3 English subset ($\approx 690\text{B}$ tokens across 616 parquet shards, 618 GiB), streamed and packed to a sequence length of 4096, tokenised with the DeepSeek-V4.1-Flash tokenizer (vocabulary size 129,281). Complementary corpora—code (UltraData-Code, $\approx 411\text{B}$ tokens, 1.22 TB), mathematics (UltraData-Math, $\approx 303\text{B}$ tokens, 552 GB), and an agentic instruction-tuning set (UltraData-SFT-Agent-2609, 51 G)—are reserved for post-training (Section V). For the full-scale production run, the larger Ultra-FineWeb base corpus (English high-quality tier, $\approx 2.66\text{TB}$ / 6,000 parquet shards, plus a Chinese subset of 256 shards / 324 GB) is being prepared to supersede L3 as the stable-phase main corpus; the L3 subset is retained for the annealing tail. Contamination control is enforced before any corpus enters training: a blacklist of 193,295 character-level 13-gram fingerprints (FNV-1a 64-bit hashed, NFKC-normalised) is built from all 536 tasks across six evaluation snapshots of EDA-Eval-PyAether, and every new data shard—including SFT corpora—is scanned for ≥ 0.80 n-gram overlap before release (Section V-D).

F. Data Mixture Proportion Search

Because the optimal web/code/math ratio is scale-dependent and a full-scale sweep is infeasible, we searched the three-way mixture at a proxy scale using Bayesian optimisation. A compact proxy model ($d=128$, $L=14$, $\approx 18.36\text{M}$ parameters—two orders of magnitude smaller than the 2.2B target and not intended as a performance predictor) was trained for each trial under Optuna’s TPESampler with MedianPruner, one trial per GPU, global batch size 2048 at sequence length 4096. The objective was the average accuracy on eight common-sense lm-eval tasks (`arc_challenge`, `arc_easy`, `boolq`, `hellaswag`, `openbookqa`, `piqa`, `sciq`, `winogrande`), each evaluated with `--limit 500` subsampling.

Two rounds of 200 trials each were completed. In the first round, using final training loss as the objective, the Spearman rank correlation between loss-ranked and accuracy-ranked mixtures was $\rho = -0.43$, confirming that loss is a weak proxy for downstream ability. In the second round, using the eight-task average directly, the top-5 mixtures were retrained and re-evaluated at the full lm-eval scale. Table III reports the results: the full-eval rankings were nearly inverted relative to the Bayesian-optimisation rankings (Spearman $\rho = -0.80$,

$n=5$, $p \approx 0.10$), and the spread across all five mixtures was only 0.383–0.390 (≈ 0.6 pp).

TABLE III

TOP-5 DATA MIXTURES FROM THE SECOND BAYESIAN-OPTIMISATION ROUND, RETRAINED AND EVALUATED AT FULL LM-EVAL SCALE. THE BO RANKING (COLUMN 5) IS NEARLY INVERTED RELATIVE TO THE FULL-EVAL RANKING (COLUMN 6), AND THE SCORE SPREAD IS ≈ 0.6 PP.

Trial	Web	Code	Math	BO	Full
23	0.941	0.109	0.010	0.4155	0.3846
38	0.895	0.100	0.010	0.4138	0.3835
118	0.874	0.071	0.055	0.4130	0.3877
199	0.931	0.098	0.010	0.4123	0.3899
33	0.940	0.086	0.010	0.4090	0.3897

At the proxy scale of 18.36M parameters, data-mixture rankings were indistinguishable within 0.6pp, leading us to adopt the 88:8:4 (web/code/math) prior from the literature rather than over-fitting to proxy-scale noise. The empirically optimal region was stable at web 0.87–0.94, code 0.06–0.12, math ≈ 0.01 , consistent with the prior. Raising the discrimination floor—via a larger proxy, multiple seeds, or full-scale evaluation replacing the `--limit 500` subsampling—is left to future work.

G. Search Budget

The pre-training search is bound by a ≈ 92 GPU-hour plan on eight H100 GPUs (≈ 192 GPU-hours wall-clock available); ≈ 85 GPU-hours were consumed. All short-horizon sweeps (S1–S4) run 5000 steps at a global batch size of $8 \times 4094 \approx 32.7$ K tokens/step (≈ 164 M tokens, ≈ 4.0 GPU-hours each); the verification long run (S5) runs 20000 steps (≈ 655 M tokens, ≈ 13.9 GPU-hours). Validation loss is reported at matched step counts so that comparisons across configurations remain apples-to-apples.

H. Results

The winning configuration is a stable learning rate of 1×10^{-3} under a warmup–stable–decay (WSD) schedule (warmup 250 steps, decay 500 steps, $\text{min_lr} = 1 \times 10^{-5}$), with an 86:10:4 blend of Ultra-FineWeb-L3, code, and math data in the annealing tail. Over the 5000-step horizon the validation loss reaches **2.7647**. The learning-rate sweep improves monotonically from 2.9037 (2×10^{-4}) to 2.7627 (1×10^{-3}); because 1×10^{-3} is also the grid maximum, that on its own leaves open whether the true optimum lies beyond the grid. Extending the grid upward closes the question: 1.5×10^{-3} gives 2.7752 and 2×10^{-3} gives 2.7889, so the sweep is *U-shaped* and 1×10^{-3} is an interior minimum rather than a clipped boundary. Each short-horizon ablation confirms a single-variable choice: WSD beats cosine (2.7679 vs. 2.8155), a 10% decay ratio beats 5% (2.7679 vs. 2.7900), a 250-step (5%) warmup beats a fixed 2000-step warmup (2.7679 vs. 2.7957), and $\text{min_lr} = 1 \times 10^{-5}$ narrowly beats 3×10^{-5} (2.7647 vs. 2.7679). Adding code and math to the annealing mix lowers the final loss monotonically (2.7621 pure L3 $\rightarrow$ 2.6793 L3+code $\rightarrow$ 2.6298 L3+code+math).

The 20000-step verification run reaches a final training loss of **2.2054**, with the decay tail (steps 19500–20000) annealing the loss from ~ 2.34 to 2.2054 (annealing gain ~ 0.14 ; Figure 1). Across three seeds the 5000-step loss is 2.6739 ± 0.0469 (mean $\pm$ sample standard deviation, $n=3$; per-seed 2.6402, 2.6539, 2.7275).

Table IV contrasts the hybrid backbone against a dense transformer at the ≈ 2 B scale, trained from scratch under matched data, seed, and GPU footprint; the hybrid offers fewer parameters and faster decoding at a modest training-throughput cost.

TABLE IV

ARCHITECTURE-LEVEL COMPARISON MOTIVATING THE HYBRID CHOICE (DENSE = MINICPM5-2B, HYBRID = BAIZE; MEASURED FROM SCRATCH UNDER MATCHED DATA, SEED, AND GPU FOOTPRINT). THE DECODE GAP IS MEASURED UNDER THE MEGATRON-CORE DIRECT-DRIVE HARNESS WITHOUT CUDA GRAPHS, WHERE THE DENSE BASELINE IS CPU-LAUNCH-BOUND, AND THEREFORE NARROWS UNDER PRODUCTION INFERENCE STACKS.

Metric	Dense	Hybrid	Edge
Parameters	2.51B	2.22B	hybrid (−11.6%)
Final loss (1k steps)	4.81	4.68	hybrid (−0.12)
Training tok/s	89K	72K	dense (+24%)
Prefill tok/s	21.4K	23.7K	hybrid (+10.5%)
Decode tok/s	2.2	22.8	hybrid (+10.3 $\times$)

Table V reports the hyper-parameter search outcomes at a matched horizon; the searched axes target the hybrid training recipe (stable learning rate, warmup length, decay ratio, and schedule family).

TABLE V

HYPER-PARAMETER AND DATA-MIX SEARCH OUTCOMES AT A MATCHED 5000-STEP (S5: 20000-STEP) HORIZON.

ID	Change vs. best	Val. loss	Outcome
S1-01	baseline ($lr=3 \times 10^{-4}$, WSD)	2.849	baseline
S2-07	stable LR = 1×10^{-3}	2.7627	best LR
S3-02	schedule family: cosine	2.8155	WSD retains
S3-03	decay ratio 5%	2.7900	10% retains
S3-04	warmup 2000 steps	2.7957	250 retains
S4-03	anneal L3+code+math 86:10:4	2.6298	best mix
S5-01	winner, 20000 steps	2.2054	winner

V. LLM POST-TRAINING

Stage (ii) adapts the pre-trained backbone to EDA scripting, and it is the stage that carries BaiZe’s domain claim: the pre-training corpus of Section IV-E is general web, code, and mathematics text and contains no EDA artifacts, so the domain prior must be introduced here. We organise the stage as three steps—*domain annealing*, *supervised fine-tuning*, and *reinforcement learning*—all evaluated under a single frozen protocol (Section V-D).

A. Domain Annealing

The cheapest lever on the domain prior is to continue pre-training on EDA text before any behavioural tuning. We

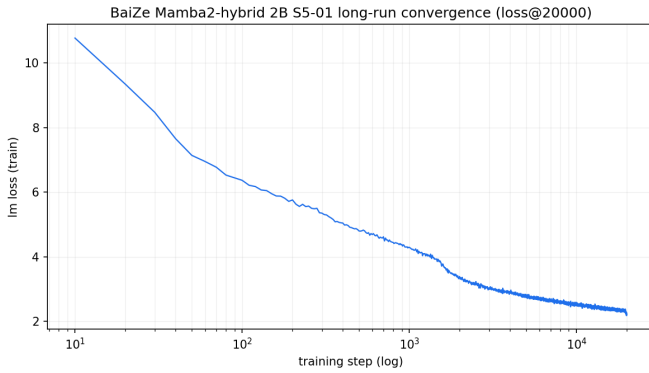


Fig. 1. Training-loss curve of the winning Mamba2-hybrid configuration over the 20000-step verification run (seed 1234); the decay tail (steps 19500–20000) anneals the loss from ~ 2.34 to 2.2054.

reuse the warmup–stable–decay machinery of Stage (i) and anneal on a corpus assembled from PyAether and SKILL API references, EDA tool documentation, and open HDL sources, blended with the general mixture at a controlled ratio. Because the step changes only the data distribution, we contrast it against an SFT-only control to isolate its contribution from the behavioural tuning that follows.

B. Supervised Fine-Tuning

SFT teaches the generate–execute–refine loop that the deployment target requires. The corpus combines the agentic instruction set (UltraData-SFT-Agent-2609, 51G across 50 shards) and a general instruction set (UltraData-SFT-2605, 319G across 1,504 JSONL files, verified byte-identical to the HuggingFace manifest) with PyAether-specific material synthesised from API documentation. We ablate the data mixture (agentic / code / mathematics / PyAether), the presence of interleaved tool-call traces, full fine-tuning versus LoRA, and the sequence length: agentic traces—code, tool returns, and successive refinements—routinely exceed the 4096-token window of Table I, so window extension is treated as a first-class variable rather than a fixed setting.

C. Reinforcement Learning with Verifiable Rewards

An execution sandbox turns correctness into a verifiable signal, so we optimise with group-relative policy optimisation and no value network. Table VI separates the reward variants we study. A binary signal is the simplest baseline; partial credit over executable attempts addresses the sparsity of pass/fail; a tool-call penalty aligns the policy with the harness cost model; and an API-legality term penalises hallucinated symbols. Because the reward is computed from assertions a policy could in principle target directly, we keep the harness anti-cheat hook enabled in every run and report a cheat rate beside every accuracy figure.

D. Evaluation Protocol

All post-training variants are scored on **EDA-Eval-PyAether** [14], a benchmark of 158 real-world EDA scripting

TABLE VI
REWARD VARIANTS STUDIED IN STAGE (II).

ID	Reward
R1	binary pass/fail
R2	executable + assertion pass ratio
R3	R2 + tool-call penalty
R4	R3 + API-legality term

tasks. Each task pairs a natural-language prompt and a function signature with assertion-based test code, and is scored by execution in the vendor sandbox: a task passes when the generated code runs without error and every assertion holds. We follow the benchmark’s own protocol so that our numbers are comparable to published ones: five independent runs with reshuffled task order and fresh context, reporting mean Pass@1 and its run-to-run standard deviation, with a 2500-second per-task timeout and a single non-restartable trace per task.

Two controls make the comparison interpretable. First, *contamination*: no task, assertion, or reference solution from the benchmark enters any training split. A blacklist of 193,295 character-level 13-gram fingerprints (FNV-1a 64-bit hashed, NFKC-normalised, with 8-gram fallback for short texts) is built from all 536 tasks across six evaluation snapshots of EDA-Eval-PyAether; every training shard—including SFT corpora—is scanned for ≥ 0.80 n-gram overlap before release, with MinHash($k=128$) tracing for source attribution. PyAether training material is drawn only from API documentation, and we isolate it by symbol so that benchmark targets cannot leak. Second, *harness parity*: only the backbone is varied. The retrieval stack, sandbox, anti-cheat hook, and agent scaffold are held byte-identical to the published configuration, because the harness supplies most of the observed accuracy and a variant harness would confound the model’s contribution with the scaffold’s.

Results are reported as [TBD].

VI. VISION ENCODER PRETRAINING

Stage (iii) pre-trains a custom vision encoder from scratch for the multimodal stage (Section VII). This section presents the architecture-selection rationale, the pre-training recipe, and the resulting measurements.

A. Architecture Selection

We compare six candidate vision encoders at an iso-parameter 500–600M scale, all *randomly initialized* (no upstream ImageNet/CLIP/SigLIP weights) so that only the architecture body is contrasted. All candidates are trained under a unified stack: `open_clip` 3.2.0 loss modules with a *frozen* CLIP-ViT-L/14-336 text tower (768-d, context 77), the CLIP InfoNCE (softmax-contrastive) objective, and the Stanford GPIC short caption split (≈ 2.77 M pairs, 0% truncation), at resolution 224 / patch 16 / bf16, learning rate 3×10^{-3} (AdamW, warmup 20), seed 1234, batch 64×8 (i.e. 512 in-batch negatives). The six candidates are:

- **OpenVision2** — a pure-attention ViT (width 1024, depth 30), 505.2M.
- **MambaEye** — a pure visual state-space model (Vim-style bidirectional SSM), 535.2M.
- **MoE-ViE** — a sparse Mixture-of-Experts encoder (222.2M active per token), 505.5M total.
- **DeepEncoderV2** — an Attention+SSM hybrid, 517.2M.
- **AIMv2** — a ViT-GELU variant, 505.5M.
- **FastViTHD** — a conv-stem + hierarchical conv/attention hybrid, 507.8M.

The last two are *iso-parameter adaptations* of the AIMv2 and FastViTHD designs to this scale, not the official released models. Table VII reports the measured parameter counts, the training loss at 3000 steps, training throughput, and—crucially—two downstream metrics on ImageNet-1k measured on the *frozen* trunk: zero-shot top-1/top-5 and linear-probe top-1.

TABLE VII

VISION-ENCODER COMPARISON (FROM-SCRATCH, UNIFIED OPEN_CLIP LOSS MODULES + CLIP INFO NCE + FROZEN CLIP-ViT-L/14-336 TEXT TOWER; STANFORD GPIC SHORT; 3000 STEPS; LR 3×10^{-3} ; 512 IN-BATCH NEGATIVES). DOWNSTREAM METRICS ARE MEASURED ON THE *frozen* TRUNK (IMAGENET-1K SELF-SPLIT, 50K VAL IMAGES), SO THEY PROBE THE LEARNED REPRESENTATION RATHER THAN AN ATTACHED CLASSIFIER; ZERO-SHOT TOP-1 CHANCE IS 0.10%. **ZS T1/T5** = ZERO-SHOT TOP-1/TOP-5 (%); **LP T1** = LINEAR-PROBE TOP-1 (%). [†] RUN TERMINATED EARLY BY THE COLLAPSE CIRCUIT-BREAKER (SEE TEXT).

Arch.	Params	Loss@3k	img/s	zs t1/t5	lp t1
OpenVision2	505.2M	4.865	3052	0.95/3.64	1.14
MambaEye [†]	535.2M	6.251	950	0.10/0.50	0.10
MoE-ViE	505.5M	5.558	1226	0.35/1.53	0.28
DeepEncoderV2 [†]	517.2M	6.251	1865	0.10/0.50	0.10
AIMv2 (adapt.)	505.5M	5.637	3393	0.38/1.61	0.25
FastViTHD (adapt.)	507.8M	5.668	2097	0.39/1.57	0.27

Finding. OpenVision2 ranks first on every quality metric—lowest loss, most dispersed features, and the best ImageNet-1k zero-shot and linear-probe accuracy (Table VII). The pure-attention family (plus MoE or a conv stem) trains normally: AIMv2 is the fastest to train but does not convert speed into quality, and neither hybrid beats a plain attention ViT at equal parameter count.

An architecture-specific failure mode. The two candidates that contain state-space sub-structure—MambaEye (pure SSM) and DeepEncoderV2 (attn+SSM)—*exactly* collapse: their loss pins at $6.25 \approx \ln 512$, the entropy of an unlearned 512-way InfoNCE problem, with a same-tower feature cosine of 1.0000 and ImageNet-1k at exactly chance (0.10%). The text tower, data, objective, batch size and seed are identical across the six runs, so the only variable is the vision architecture; under this contrastive recipe the collapse is tied to the SSM sub-structure rather than to the recipe. We report it as a negative result, with its boundary stated: an earlier sweep under a different objective trained MambaEye without collapsing, so this is a property of the (architecture, objective) pair rather than evidence that state-space vision encoders are

unusable.

Methodological note. An earlier sweep at $lr=10^{-3}$ pinned every ablation cell at an identical loss, because the cosine schedule collapsed the learning rate to $\approx 10^{-5}$ before the run ended. We therefore re-ran all configurations at $lr=3 \times 10^{-3}$; sub-1k-step comparisons, and any 3000-step comparison at $lr=10^{-3}$, are uninformative, and every number in this section uses the corrected rate.

B. Scaling Behaviour

Holding the winning architecture fixed, we ask how far the representation improves with data, and how tower width trades against data at a fixed sample budget. Stage-1 trains three widths for 30000 steps on the same ≈ 18.5 M-pair corpus (CC12M + Amshaker) under the recipe above: linear-probe top-1 falls monotonically with width—6.10% (w512, 126.8M), 3.63% (w768, 284.5M), 1.03% (w1024, 505.0M). At this data scale the *smaller* tower is therefore the more sample-efficient one. Stage-2 then trains w512 for 108,000 steps (55.3M samples) and evaluates every 10,000-step checkpoint on the frozen trunk (Table VIII): linear-probe top-1 rises from 3.96% at 5.1M samples to a peak of 7.70% at 51.2M.

TABLE VIII

DATA-SIDE SCALING OF THE SELECTED VISION TOWER (OPENVISION2, w512 = 126.8M, FROZEN TRUNK, IMAGENET-1K LINEAR PROBE AT 10,000-STEP CHECKPOINTS).

Samples (M)	5.1	10.2	20.5	30.7	41.0	51.2	55.3
lp top-1	3.96	4.39	6.46	7.35	7.30	7.70	7.40

A power law $acc = 0.251 - 0.864 N^{-0.090}$ fits the eleven checkpoints ($R^2 \approx 0.94$) and implies an asymptote of 25.1% for the pure contrastive (InfoNCE) objective: even with unlimited data this from-scratch contrastive route saturates near 25% linear-probe top-1, and reaching 20% would take on the order of 6×10^{10} samples under the more favourable log-linear extrapolation—roughly three orders of magnitude beyond the $\approx 1.2 \times 10^8$ unique image-text pairs available locally. A complementary five-point model-size sweep (widths 384–1024 at matched samples) finds the marginal effect of width to be negative throughout (≈ -2.2 to -2.4 linear-probe points per doubling of parameters) with no saturation point: the best configuration observed is the smallest tower tried (71.5M, 7.99% at 15.4M samples). We initially read this as the central limitation of the vision stage—the binding constraint is data volume and objective function, not architecture. The next subsection revisits that reading: the ceiling is specific to the contrastive objective, and switching to a caption-independent dense objective while retaining the contrastive term breaks it.

C. Breaking the Contrastive Ceiling with Dense Supervision

The 25.1% asymptote above is tied to a single global scalar signal per image-text pair (the InfoNCE loss). A body of recent work argues that *dense per-patch* supervision can be more sample-efficient than global contrastive learning:

AIMv2 [15] reports “denser supervision compared to discriminative objectives” from a masked-patch generative head, SigLIP 2 [16] augments sigmoid contrastive loss with captioning and masked-prediction heads, and OpenVision 2 [17] removes the text encoder entirely in favour of a purely generative objective. Motivated by these findings, we test whether augmenting the contrastive loss with a caption-*independent* dense term can break the 25.1% ceiling without additional data.

a) *AIMv2-style objective.*: We add a masked-patch reconstruction head to the selected tower: 60% of patches are masked and their raw pixels are reconstructed by a lightweight predictor (0.66 M params) via an MSE loss, while the InfoNCE contrastive term is retained at equal weight ($\lambda_{\text{patch}} = 1.0$). The dense term is *caption-independent*—it supervises raw pixels, not text tokens—so it does not depend on caption quality or truncation. This is a self-developed adaptation of the AIMv2 design philosophy, *not* an official reproduction: the official AIMv2 [15] uses prefix-LM attention, $\alpha \approx 0.4$ loss weighting, and ~ 12 B training pairs, none of which we replicate here.

b) *Result: the ceiling is broken.*: Under the *identical* architecture (OpenVision2, w512), data (≈ 18.5 M-pair corpus, CC12M + Amshaker), and optimisation schedule as the scaling study above, the AIMv2-style objective produces a monotonically increasing linear-probe trajectory that exceeds the contrastive baseline at every checkpoint (Table IX): lp top-1 rises from 11.50% at 5.1 M samples to **19.76%** at 55.3 M—already above the 7.40% contrastive baseline at the same sample budget by +12.4 pp, and the gap *widens* monotonically from +7.5 pp at 10.2 M to +12.4 pp at 55.3 M. A power-law fit $acc = a - bN^{-c}$ yields $R^2 = 0.91$; the exponent $c = 0.040$ is much smaller than the contrastive-only fit ($c = 0.090$), so the curve saturates far more slowly and no asymptote can be identified within the observed range. A subsequent extension to 176 M samples (3 epochs over the full 58.8 M-pair corpus, 344 k steps) reaches 20.27%, still rising with no saturation signal; a 35-checkpoint refit gives $R^2 = 0.94$, but the power-law asymptote hits its upper bound ($a \rightarrow 1.0$) and the trajectory is non-monotonic (7 of 34 checkpoints decrease, attributable to resume transients), so the precise ceiling remains indeterminate.

TABLE IX

AIMV2-STYLE OBJECTIVE VS. CONTRASTIVE BASELINE (OPENVISION2, w512 = 126.8 M, FROZEN TRUNK, IMAGENET-1K LINEAR PROBE AT 10,000-STEP CHECKPOINTS). THE DENSE OBJECTIVE EXCEEDS THE CONTRASTIVE BASELINE AT EVERY CHECKPOINT, WITH THE GAP *widening* AS DATA GROWS. $\Delta = \text{LP}_{\text{AIMV2}} - \text{LP}_{\text{InfoNCE}}$ (PERCENTAGE POINTS).

Samples (M)	5.1	10.2	20.5	30.7	41.0	51.2	55.3
InfoNCE lp	3.96	4.39	6.46	7.35	7.30	7.70	7.40
AIMv2-style lp	11.50	12.73	15.33	16.67	—	18.84	19.76
Δ (pp)	+7.5	+8.3	+8.9	+9.3	—	+11.1	+12.4

c) *Causal decomposition: both terms are necessary.*: A four-arm ablation at 15.4 M samples localises the effect:

- **Arm (i)** InfoNCE baseline: 6.08%.
- **Arm (iv)** CoCa (contrastive + *caption-dependent* generative CE): 0.47%—the caption-CE term collapses the frozen trunk, confirming that *caption-dependent* generative supervision is harmful in this regime.
- **Arm (vi-B)** AIMv2-style without the contrastive term (pure masked-patch AR): 6.23% (+0.15 pp over baseline—no breakthrough).
- **Arm (vi-A)** AIMv2-style with both terms: 12.08% (+6.00 pp—breakthrough).

The breakthrough therefore depends on the *interaction* of the contrastive anchor and the caption-independent dense term: neither alone suffices. Removing the contrastive term (arm vi-B) eliminates the breakthrough, and replacing the caption-independent term with a caption-dependent one (arm iv) actively collapses the representation. This is consistent with MAE [18], where pure masked-image modelling yields lower linear-probe accuracy than contrastive methods, and with AIMv2 [15] and SigLIP 2 [16], which retain a contrastive or sigmoid-contrastive signal alongside the dense head.

d) *Boundary.*: All AIMv2-style results are from a self-developed adaptation (not the official AIMv2), trained on ≤ 58.8 M unique pairs (vs. ~ 12 B for the official model [15]). The precise asymptote of the dense route remains indeterminate: a 35-checkpoint scaling refit over 5–176 M samples gives $R^2 = 0.94$, but the power-law parameter hits its upper bound ($a \rightarrow 1.0$) and the trajectory is non-monotonic (7 of 34 checkpoints decrease, attributable to resume transients), so we report it honestly without forcing an extrapolation. Nonetheless, 176 M samples show no saturation signal—the dense route reaches 20.27%, a level the contrastive route would need on the order of 10^{10} samples to attain—confirming that the 25.1% ceiling is specific to the contrastive objective. The full-data mix (GPIC + CC12M + Amshaker, 58.8 M pairs) is less sample-efficient at matched N than the CC12M+Amshaker-only corpus (16.80% vs. 19.76% at ≈ 55 M samples), likely because GPIC long and tag captions are truncated at the 77-token context; the *caption-independent* patch-MSE term is unaffected, and the dense route still exceeds the contrastive baseline by a wide margin (+9.4 to +12.4 pp at ≈ 55 M samples, depending on data mix). The four-arm decomposition is conducted at a single sample budget (15.4 M); the interaction effect (both terms necessary) is confirmed at larger scale, where the full AIMv2-style objective continues to exceed the contrastive baseline by a wide margin through 176 M samples, while removing the contrastive term (arm vi-B) yields no breakthrough at any scale tested.

D. Configuration and Ablations

Holding the winner (OpenVision2) fixed, we sweep the core degrees of freedom single-variable-at-a-time (3000 steps each, at $lr=3 \times 10^{-3}$ —the rate at which the schedule does not collapse, see the note above): resolution/patch (Table X), objective function (Table XI), and learning rate, plus a multi-seed robustness check and a held-out retrieval proxy.

TABLE X

RESOLUTION/PATCH ABLATION ON OPENVISION2 UNDER THE CORRECTED OBJECTIVE (RUN ON THE LLaVA IMAGENET/EN 500K ARM, HENCE A DIFFERENT LOSS SCALE FROM TABLE VII, WHICH USES GPIC), ALL RUNS 3000 STEPS. 224/16 IS BEST ON BOTH AXES: LOWEST LOSS AND HIGHEST THROUGHPUT; THE COARSEST GRID RETAINS ONLY $\approx 56\%$ OF THAT THROUGHPUT, SO ALL LATER STAGES USE 224/16.

res/patch	Val. loss
224/16	3.0977
336/14	3.1909
448/14	3.2711

TABLE XI

OBJECTIVE FUNCTION AND LEARNING RATE ON OPENVISION2, ALL RUNS 3000 STEPS. THE TWO OBJECTIVES PRODUCE LOSSES ON DIFFERENT SCALES AND ARE *not* COMPARABLE; THE ONLY COMMENSURABLE QUANTITY IS THROUGHPUT, WHERE THEY TIE TO WITHIN 1%. LEARNING RATE, BY CONTRAST, IS DECISIVE: THE ORIGINAL ANCHOR COLLAPSED THE SCHEDULE TO ITS FLOOR.

Variant	Loss	img/s	Note
SigLIP	4.4562	1645	sigmoid
CLIP InfoNCE	3.2704	1657	softmax; scale differs
lr 10^{-3}	4.9962	1651	schedule collapsed
lr 3×10^{-3}	4.4562	1645	selected
lr 5×10^{-3}	4.4542	—	—

a) *Evaluation.*: Domain task metrics are not yet available for this stage, so we adopt the standard frozen-trunk protocol: ImageNet-1k zero-shot (class prototypes synthesised from the frozen CLIP text tower) and a linear probe on the frozen trunk, over a deterministic self-split of the ImageNet-1k training set (50 images per class for the probe and 50 for validation, disjoint from every corpus used for training). We first tried a zero-shot text $\leftrightarrow$ image retrieval proxy ($R@1/5/10$ on a held-out 5K subset of laioncn/EN, disjoint from the training source) and found it *degenerate*: $R@1=0.0002$, exactly chance for 5000 candidates, separating neither learning rates nor architectures. We report that as a methodological warning and use ImageNet-1k instead, on which the architectures do separate—healthy candidates reach $2.5\text{--}9.5 \times$ chance, whereas the two collapsed ones sit at exactly chance. Resolution is not a free variable in this comparison: 224/16 is best on both loss and throughput (Table X). Stage (iv) revisits the visual-token count for detail-dense layout and schematic views, where the reasoning of this section—that a 196-token view is dominated by coarse semantics—does not transfer.

VII. MLLM ALIGNMENT

Stage (iv) connects the vision encoder of Section VI to the language backbone through a lightweight projector, following the staged curriculum established for open multimodal assistants: projector pre-training, mid-training, and instruction tuning. The design is motivated by the observation that the dominant EDA artifacts are visual—layout and schematic views account for 70.2% of the tasks in the evaluation bench-

mark [14]—so a multimodal core can be scored on the same execution-based protocol as the text-only one.

a) *Task construction.*: We build three task families, each chosen because it admits an objective ground truth. (i) *image $\rightarrow$ script*: given a rendered layout or schematic view, produce the PyAether or SKILL script that recreates it; super-vision is available because executing a script yields its own reference rendering. (ii) *script $\rightarrow$ image*: predict the geometry that a given script produces, again self-labelled by execution. (iii) *layout VQA*: answer attribute questions—layer identity, device type, spacing and rule violations—labelled by domain experts. The first family is the strongest of the three: its output is a script, so it is scored by the same assertion sandbox used in stage (ii), which gives the multimodal and text-only stages a common metric instead of two incommensurable ones.

b) *Curriculum and ablations.*: We run stage-1 projector pre-training on caption-style pairs, stage-1.5 mid-training on the three families above, and stage-2 instruction tuning on the full mixture. The ablations vary the projector (MLP versus query-based), whether the vision tower is frozen or updated, and the number of visual tokens per image. The last is of particular interest. The architecture study of Section VI selected 224/16, where an image becomes only 196 tokens—a regime in which the training objective is dominated by coarse semantics; whether that setting transfers to detail-dense views is precisely what stage (iv) tests. Layout and schematic views are the opposite: they are detail-dense line drawings whose meaning lives in fine geometry. We therefore treat resolution and visual-token count as first-order variables here rather than inheriting the earlier null result, which we read as a property of the proxy task and not of the encoder.

Results are reported as [TBD].

VIII. SEMICONDUCTOR DOMAIN ADAPTATION

Stage (v) completes the pipeline by adapting the multimodal model to proprietary design practice. It differs from stage (ii) in two respects, which we state explicitly because otherwise the two stages would coincide: the reward is no longer verifiable but *learned from human judgement*, and the target behaviour is no longer general API usage but a team’s internal conventions—naming rules, layer policies, review checklists—which no public benchmark measures.

a) *Method.*: We collect pairwise judgements from in-house design reviews over the stage-(iv) task families, fit a domain reward model on them, and optimise the aligned model against that model under a KL-regularised objective. Because preference data is far scarcer than the verifiable signal of stage (ii), we initialise from the stage-(iv) checkpoint and constrain the update rather than training the objective from the pre-trained model.

b) *Why a small model.*: The rationale for a $\approx 2.2\text{B}$ core is strongest here, and it is a constraint rather than a preference. Proprietary layouts and proprietary preference data cannot leave the premises, so the choice is not between a large hosted model and a small local one—it is between a small local model and no model at all. Serving cost then dominates, and for an

agentic session the relevant cost is not throughput on a short prompt but memory that grows with accumulated context: in the hybrid backbone of Section III-A, the four attention layers are the only component whose serving state grows with the session, while the Mamba-2 layers hold a constant-size recurrent state.

c) Evaluation.: We evaluate on held-out review items using expert agreement, and report the result beside the execution-based numbers of stage (ii) so that verifiable and preference-based adaptation can be compared on the same system rather than in separate accounts.

Results are reported as [TBD].

IX. CONCLUSION

We presented BaiZe, an end-to-end multimodal agentic system: a $\approx 2.2\text{B}$ -parameter hybrid (state-space + attention) language model built from scratch, post-trained for agentic capability, paired with a custom vision encoder, aligned into a multimodal model, and adapted to the semiconductor domain. Through a controlled from-scratch comparison, we selected a sequential Mamba2-hybrid backbone over a dense transformer for its lower loss-per-parameter and substantially faster inference, then applied a staged, budget-constrained hyperparameter search (S1–S5) on the final backbone. Stages (ii) to (v) are specified here as a protocol and evaluated on a common execution-based benchmark [14]; their measurements are reported as [TBD]. The complete recipe and a single reproducible command encapsulate the entire construction under the NVIDIA NeMo recipe.

The immediate next step is agentic fine-tuning with the UltraData-SFT-Agent corpus toward a deployable agent core for EDA and embedded applications. We hope this blueprint lowers the barrier for cost-bounded teams building their own efficient agentic SLMs from scratch.

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